

Lean Six Sigma Case Study Project

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Integrating the Concept of Lean in a Healthcare Environment

The concept of lean can be integrated into a healthcare system through top-down management. For example, a group of higher-ups can observe an interaction between a customer and a medical specialist, note all the potential errors in the interaction, and look to improve upon the interaction by gradually eliminating the errors (3). In doing this, the higher-ups have already identified areas where the interaction could be improved and have effectively facilitated the implementation of lean. As more observations are made, more data can be gathered and recorded to discover the main sources of error between the interactions. After the data is gathered and analyzed by a team of researchers, possible hypotheses can begin to form about the sources of error. For example, the interaction between the customer and the doctor could be better if there were fewer unnecessary movements, shorter wait times, or lower levels of inventory to search through. Then, after the hypotheses have been made, possible improvements could be made and sustained for long-term benefits. Once employees grasp a better understanding of the new processes, they will be able to better eliminate waste in the interactions and boost customer satisfaction.

Another way lean can be integrated in a healthcare system is through practicing a culture of continuous improvement (5). To promote a culture of continuous improvement, the healthcare system should alert all employees about the standardized process they have developed through research (3). If workers and employees can follow the standardized process, they will guarantee that they are gradually eliminating waste and improving. For example, the healthcare system could have a sheet of paper that dictates the correct procedures to be followed by the medicine prescriber. After the worker does his or her job, he or she can confirm whether they followed the standardized procedure. Through repetition, the worker will also be able to learn the procedure faster and become better. For example, a medical advisor that has no experience prescribing drugs to patients can follow a standardized procedure and eventually become better at doing their job.

Lean can also be implemented in a variety of other ways in a healthcare system. For example, setting up surveillance televisions that alert workers when there is a medical emergency can be crucial to making sure that patients are treated as quickly as possible to avoid fatal errors. Another way to incorporate lean could be by looking at the healthcare system from a different perspective. For example, a doctor or nurse could hypothetically put themselves in the shoes of the customer and then try to make the process as painless as possible by figuring out what drive customer satisfaction. For instance, maybe the wait times are making the customer feel

uncomfortable before they are operated on or prescribed medication (3). A successful healthcare system will see its image in the eyes of the customer and look for ways to reduce the waiting time. Furthermore, maybe the amount of paperwork the customer must fill out is overbearing (3). In response to this, the healthcare system can look to reduce the amount of paperwork the customer must complete.

The Importance of Lean Integration in American Healthcare

The integration of lean practices into a healthcare system is paramount to long-term success in today's environment for a variety of reasons. Lean is designed to take often complex processes and boil them down into more easily understood standardized processes that encourage sustainability and automation (4). By doing so, a company can begin to see a gradual reduction in wasteful activities within processes. This reduction in wasteful activities will then lead to both better patient experiences and outcomes. This is important as the healthcare system is charged with the responsibility of ensuring a patient's health in safety.

Standardized processes created through lean thinking can increase patient satisfaction with the hospital and its staff. In the process of just seeing a physician there are many possible wastes; incomplete paperwork, exam rooms being excessively far from supply closets, etc... (8). By replacing these wasteful activities using lean methodology we can improve patient experiences, and also reduce costs as hospital staff will not be tied up constantly with these wasteful activities.

Ensuring that processes in healthcare are lean optimized, Providers can also reduce the number of defects or errors that can occur (4). With optimized procedures, staff can focus more on delivering high quality and professional healthcare to patients. For example, though even routine surgeries carry some amount of risk and can have complications, having a standard operating procedure that can apply to a majority of patients can prevent new and even veteran surgeons from making needless mistakes, costing a patient their life. It also allows for more patients to have access to the Operating Room (O.R.) as surgeons are not wasting O.R. time determining if something falls within hospital procedure.

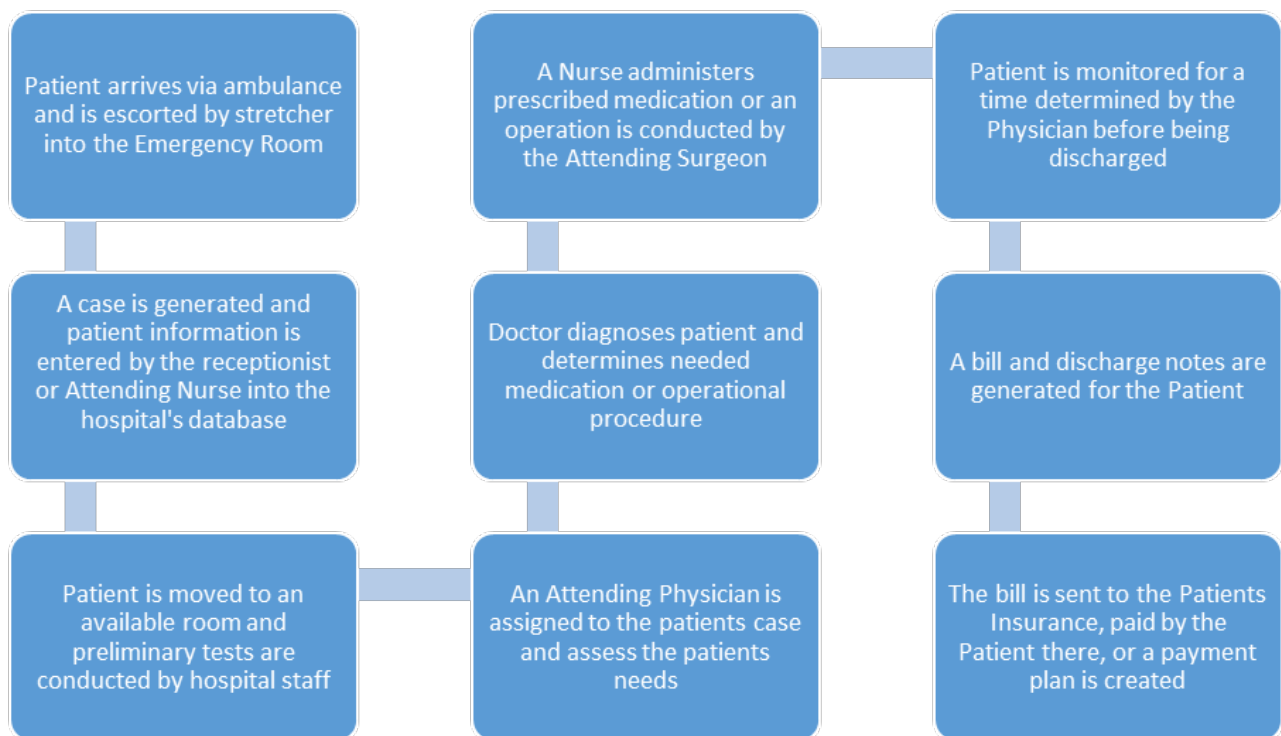
Lean also uses what is known as a "bottom-up" approach in its methodology (4). Staff that work hands-on in healthcare will have an input in the new processes. Including general staff in the creation of these new processes and procedures will help foster a healthy work environment where they will be more willing to actively search for errors in their day to day duties. When staff are involved and know that their input is valued and considered their happiness will increase. Research suggests that these happy employee's will actually begin to outperform their less happy counterparts leading to a more efficient healthcare system (9).

Continuous improvement is another major factor in lean. The healthcare field is ever expanding with new procedures and medications, so Providers in turn must be constantly expanding their knowledge (10). This constant focus on educating staff leads to better overall

patient care and staff happiness (9). As well, a more educated workforce can also spot errors in processes that may normally go unnoticed.

Healthcare Providers that fail to implement lean, leave themselves vulnerable to errors within their processes. A key point in lean is data collection. This data collection and analysis is what allows for the discovery of these errors. Failing to collect the data and analyze it could lead to these errors becoming large enough to substantially impact the Provider's patient care. Decreasing patient happiness with the Provider and leading to a reduction in both patients, and profits. As long as patients are happy and receive good healthcare services, they are more likely to return to that Provider.

Process Map for a Patient Entering the Emergency Room



(Alexandre Evdokimov BS RT (N) (CT), personal communication)

Inputs = Patient Arrival

Process = Case is generated -> Patient is moved to a room -> Physician is assigned to the case -> Patient is diagnosed -> Prescription is ordered, or surgery is scheduled -> Medication is administered, or surgery is conducted -> Patient is monitored

Output = Bill and discharge notes are generated -> Patient is billed and discharged

DMAIC Concept Interpretation

A common problem found in emergency rooms is excessive wait times. In a study conducted by Keith A. Willoughby, they found that on average patients would wait 112 minutes for a consultation (7). The study more acutely describes how a large portion of these wait times can be attributed to patients awaiting consultation from a Physician. Using the DMAIC process, we can analyze why these patient consultations take such a long time.

First, we define what the problem is. The study notes how there isn't a clear link between what Doctor is assigned to which patient (7). Without a clear designation of what Doctor is assigned to which patient, staff may have to waste time trying to identify that patient's Doctor. This leads to longer wait times to diagnose patients, request prescriptions, treatments, and discharge patients.

Now that the problem has been identified, it needs to be measured. This is an integral part of the process as without any measurements it becomes impossible to discern whether improvements in the process made any difference. Of the five hospitals included in the study, the average wait time for a consultation was 112 minutes (7). After the measurement process, an analysis of the data is conducted, and it is found that the average transit time was nearly five hours for the five hospitals in the study (7). The analysis also showed that the total patient transit time for the lowest acuity patients ranged up to two hours, while the highest acuity patients ranged up to six-and-a-half hours (7). Overall, the analysis found that out the total patient transit time, approximately one-half is spent waiting with consultation taking up most of the waiting time (7).

The RUH team involved in the study conducted a series of three PDSA cycles with different ideas on how to reduce the reassessment time. Some of the improvement ideas in these cycles were to create a color code for each patient for the responsible physician and training the physician to identify their own patients with colored dots, creating a visual reminder on the whiteboard for the physician to see which patients is ready for assessment, and a Trial of "Think Reassessment" worksheet for the RNS to know when the appropriate time to ask the MD to reassess the patients (7).

Lastly, once the improvements have been made, the hospital should seek to "control" its improvements. By controlling the improvements, they will be able to maintain the benefits of the improvements in the long run. Some suggestions for controlling these improvements would be to have periodic tests and another DMAIC review of the system to ensure that the new process is functioning to standard. As well, the tests can help determine if there are other methods that can also be used to improve the process.

By defining the issues, measuring and gathering data on the issues, analyzing the data gathered, creating improvements based on the analysis, and controlling the improvements the healthcare system makes, the healthcare system can find more efficient ways to provide patient care.

Effective Principal #12

To implement lean within the healthcare system, principle 12 should be practiced and emphasized. "Go and see for yourself to thoroughly understand the situation" (2) Principle 12 is one of the most important principles when it comes to implementing lean because it states that people should "go and see" how the work is done (2). Studies have proven that people who go and visualize how the process works will have a much better understanding of how it works compared to those who have not seen the process (2).

For example, someone that has seen the process will be able to understand if the workers are following the standardized procedure in their work, if the process is operating at a smooth flow, and if the process is implementing "just-in-time" by getting their products and services delivered just before they are needed (2). Furthermore, by using "Genchi Genbitsu," otherwise known as "go and see," the worker or stakeholder could see information that is not necessarily provided on papers or data (2). If a problem happens in the process, the information on the papers can only tell so much. Therefore, by going and seeing, one can better understand the context of the problem and have a better solution.

However, the process of "going and seeing" is more complex than just seeing how the process works (2). "Going and seeing" is about physically going to the place where the work is done, asking several sophisticated questions, and analyzing the data that you observed (2). One thing is to see how the process works, but another thing is to be able to generate questions and analyze data with an understanding of how the data can be used to determine the correct way to proceed (2). For example, some possible questions that could be asked are, "What do you see?" "What are the issues?" and "What happened in the process?" (2). After asking these questions, one can better internalize what they have seen rather than just seeing it.

Then, the person who runs the analysis should be able to see where the data is coming from in terms of statistics and variables. For example, a person that is only told that a variable "x" is causing a process to not run well and contributing to overall error will not handle the data as well as a person that has gone and seen what the process is and understand what variable "x" is. Therefore, one who has internalized the process they saw will be able to directly correlate which data or statistic came from where. In the context of a healthcare system, a manager or higher-up could observe the process of a patient being operated on and take notes on the procedure of the operation, potential wastes, and meaningful variables that should be considered.

Individuals who use "Genchi Genbitsu" on a regular basis will be able to implement a culture of continuous improvement within the process, as well as better understand the root cause of error and the data (2). Managers or anyone who observes their company's processes can undoubtedly identify details that contribute to waste. For example, if a manager notices a process requires more movement between the time a worker goes to pick up their equipment and when they make their product. The manager can then spot this source of wasted time and change the

location of the equipment to make it easier for the workers. After making it easier for the workers to gather their equipment and continuously looking for more ways to improve the process, the data will reflect these improvements by showing a higher rate of productivity and efficiency. Furthermore, after such improvements are made through "Genchi Genbutsu," the managers or observers should be constantly looking for more ways to improve their process so that they can continue to exhibit "kaizen" (continuous improvement) for their company (2).

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